C

ACT SUBJECT TEST — MATH 2 • UNIT 3.01

Sequences, Series & Binomial Theorem

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Sequences, Series & Binomial Theorem Practice Exercises

1 MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

A 131
B 138
C 140
D 145

2 MCQ

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where k is a real constant. What is the first term, a_1, of this sequence?

A 6
B 10
C 19
D 26

3 MCQ

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 \neq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

A 64
B 96
C 144
D 192

4 MCQ

The bar chart below shows the values of the first four terms of an arithmetic sequence, a_1, a_2, a_3, and a_4, where $a_1 = 6$, $a_2 = 11$, $a_3 = 16$, and $a_4 = 21$. What is the value of the 8th term, a_8, of this sequence?

A 36
B 41
C 46
D 48

5 MCQ

The first three terms of a geometric sequence of positive real numbers are $k - 1$, $k + 2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

A 24
B 48
C 96
D 192

6 MCQ

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

A $\frac{256}{13}$

B $\frac{128}{13}$

C $\frac{64}{7}$

D 16

7 MCQ

The first four terms of an arithmetic sequence are shown below.

$$4, 11, 18, 25, \ldots$$

What is the 31st term of the sequence?

A 210

B 214

C 217

D 221

8 MCQ

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

A 155

B 160

C 315

D 630

9 MCQ

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k - 1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum

$$\sum_{k=1}^{\infty} a_{3k-1} = \frac{3}{13}S.$$

Furthermore, the infinite series formed by the squares of the terms has sum

$$\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S.$$

What is the value of $\sum_{n=1}^{\infty} na_n$?

A $\frac{9}{2}$

B $\frac{27}{4}$

C $\frac{81}{4}$

D $\frac{27}{2}$

10 MCQ

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, a_{10}?

A 52
B 56
C 64
D 58

11 MCQ

A geometric sequence has $a_1 = 3$ and common ratio $r = 3$. Find a_3.

A 81
B 27
C 30
D -27

12 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = -3$. Find a_3.

A 51
B -54
C -162
D 54

13 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find a_6.

A -66
B 64
C -64
D 128

14 MCQ

A geometric sequence has $a_1 = 4$ and common ratio $r = 2$. Find a_3.

A 16
B 32
C 18
D -16

15 MCQ

In an applied problem, a geometric sequence has $a_1 = 4$ and common ratio $r = -3$. Find a_5.

A 321
B -324
C -972
D 324

16 MCQ

Suppose a geometric sequence has $a_1 = 3$ and common ratio $r = -2$. Find a_4.

A 24
B 48
C -26
D -24

17 MCQ

An observed geometric sequence has $a_1 = 5$ and common ratio $r = -2$. Find a_4.

A -42
B -40
C 40
D 80

18 MCQ

Consider a geometric sequence has $a_1 = 5$ and common ratio $r = -3$. Find a_5.

A 402
B 405
C -405
D -1215

19 MCQ

In an applied problem, a geometric sequence has $a_1 = 3$ and common ratio $r = 2$. Find a_4.

A 24
B 26
C -24
D 48

20 MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -2$. Find a_6.

A -32
B -34
C 32
D 64

21 MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -2$. Find a_5.

A 16
B -32
C 14
D -16

22 MCQ

Suppose a geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_6.

A 489

B -486

C 486

D 1458

23 MCQ

An observed geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_3.

A -18

B 54

C 18

D 21

24 MCQ

Consider a geometric sequence has $a_1 = 3$ and common ratio $r = 2$. Find a_6.

A -96

B 96

C 192

D 98

25 MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = -3$. Find a_3.

A -135

B 42

C 45

D -45